# COGNITIVE MODELLING
# RAGALENS

interpretable pitch-based pipeline with strict
track-level evaluation

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# PROGRESS UPTO WEEK 3
# LAST POINT OF DISCUSSION

**RECAP**

Integration of SARAGA, Preprocoessing Scripts, Minute Front-end design

**WEEK 3**

- Pitch extraction upgraded: librosa pyin → torchcrepe
- Tonic estimation validated against annotations, began tracking octave error failure mode
- Swara mapping module initiated and stabilised
- Repo Cleaned

**WEEK 4**

lLibrosa to TorchCrepe,  Tonic Estimation Valildation increase, Swara Mapping Module

- Feature extraction implemented and refined:
- Transition matrix features
- N gram features
- Pitch class distribution features
- MFCC baseline as comparison
- Training and evaluation pipeline established with GroupKFold by track
- Results generation scripts and report figures now being updated

**PLAN FOR WEEK 5**

Feature Extraction, Finalization of final features, Updating Report

# CURRENT PROBLEM FORMULATION
# **CHANGE IN SCOPE OF PROJECT**

Core Technical Question: Can we recognise rāga identity from pitch-based symbolic representations?

**Computational View**

Pitch contour (F0) → tonic (Sa) → swara mapping → sequence features → classifier

Why pitch-based

- Avoid reliance on timbre/instrumentation
- Target melodic grammar: swara transitions + phrase structure

**Cognitive View**

Pitch contour (F0) → tonic (Sa) → swara mapping → sequence features → classifier

- Establish tonic reference from drone and context
- Categorise continuous pitch into discrete swaras
- Recognise characteristic phrases and transitions
- Accumulate evidence over time before deciding a rāga

# FEATURE SETS, **DATASETS**



## How I’m currently doing them:

### Balance

- SARAGA dataset is highly imbalanced: many segments in some rāgas, very few in others
- Segment quality control via voiced ratio filtering:
- Before filtering: 4487 segments
- After filtering (voiced ratio ≥ 0.15): 2453 segments

### Evaluation

- Keeps all segments from the same recording in the same fold
- Prevents leakage and tests true generalisation to unseen performances
- Produces honest accuracy estimates

### Feature Sets

- Feature sets compared
- Transition matrix features (swara bigram transitions)
- Pitch class distribution features
- MFCC features (timbre baseline)
- N gram features
- New: transition plus auxiliary (token rate, entropy, glide ratio)

# EXPERIMENTAL CONDITIONS **CHANGE IN RESULTS**

## 2 Experiments

- 15 rāgas: realistic but harder and more imbalanced
- 9 rāgas: well supported subset used as a stable baseline condition

Example headline numbers

- 15 rāgas: SVM transition approx 48.7% segment, 55.9% track
- 9 rāgas: SVM transition approx 58.5% segment, 69.2% track

## Feature Sets

- Transition features outperform every other feature family consistently
- MFCCs perform worst

- As class support improves, generalisation improves
- Transition features capture rāga grammar more effectively than pitch distribution or timbre

## Per Raga Breakdown

- Bhairavi: ~100%
- Tōḍi: ~92%
- Bēgaḍa: ~86%
- Mōhanaṁ: ~75%
- Ṣanmukhapriya and Rītigauḷa: ~67 to 80%

- Some rāgas are learned robustly even under strict track level validation
- Remaining errors likely reflect similarity and training data limits

This supports the claim that the model is learning melodic identity, not just memorising.

Under strict track-level validation, melodic grammar features (swara transitions) consistently outperform pitch distribution and timbral features, suggesting that rāga identity is primarily encoded in sequential pitch structure rather than surface acoustics.

Instead of treating rāga recognition as generic classification, I propose:

# INTERPRETABLE **PITCH BASED PIPELINE** AS A **COGNITIVE MODEL**

- Each stage maps to a perceptual or cognitive step
- Errors can be interpreted as meaningful musical similarity rather than mere failure

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# EXPERIMENTAL CONDITIONS **CHANGE IN RESULTS**

**Cognitive Model Understanding**

- Interpretable pitch based pipeline as a cognitive model
- Each stage maps to a perceptual or cognitive step
- Errors can be interpreted as meaningful musical similarity rather than mere failure

**Feature Model Comparison**

- Transition Features > PCD > MFCC
- Transition features capture melodic grammar
- PCD captures static tonal distribution
- MFCC captures timbre — fails under grouped evaluation
- MFCC failure indicates that timbral similarity does not encode rāga identity under cross-performance validation.

**Prior Work**

- Many prior rāga recognition papers use:
- random segment splits
- non grouped evaluation
- leakage between train and test
- This work uses:
- strict GroupKFold track level evaluation
- segment and track level metrics
- interpretability throughout the pipeline

# ANALYSIS OF **ERROR**

## Confusion Matrix

- Interpretable pitch based pipeline as a cognitive model
- Each stage maps to a perceptual or cognitive step
- Errors can be interpreted as meaningful musical similarity rather than mere failure

## Case Studies of Tracks

- For misclassified tracks, examine:
- voiced ratio quality; 0.10 too noisy, 020 strict
- tonic estimation error or octave error
- segment vote confidence
- which rāga it confuses with and why
- +7–13% improvement when moving from segment to track voting; Rāga identity emerges from cumulative phrase evidence rather than isolated short windows.

## Ablation Studies

- ground truth tonic vs estimated tonic
- voiced ratio threshold 0.10 vs 0.15 vs 0.20
- segment length variation
- transition vs transition plus auxiliary
- SVM transition vs LSTM sequence model

# GUIDANCE AND **QUESTIONS**

| | |
|---|---|
| **Scope Change** | Is the cognitive modelling and interpretability framing appropriate as the main thesis story? |
| **Cognitive Modelling Evidence** | Would it strengthen the dissertation to formally quantify swara overlap between rāga pairs and correlate it with confusion matrix values? |
| **Failure analysis structure** | Is it appropriate to include 3–5 detailed failure case studies showing tonic error, low voiced ratio, or melakarta similarity? |
| **Ablation priority** | Of tonic GT vs estimated, voiced ratio threshold, and segment length, which two ablations would you prioritise for depth over breadth? |
| **Accuracy interpretation** | Given that strict GroupKFold reduces accuracy compared to prior literature that uses random splits, is methodological honesty and interpretability sufficient even if absolute accuracy is lower? |

**I’m aiming to position accuracy as one outcome, but my main dissertation strength is strict evaluation and interpretable findings. I want to check if this is acceptable for a first class grade even if the absolute numbers are lower than papers that use random splits.**